

Correction to the `D0_corr` Measurement in CT-HYB

1 Introduction

The `D0_corr` measurement is designed to reconstruct the density-density correlation function $Q(\tau) = \langle n(\tau)n(0) \rangle$ from the dynamical weights of the Lang-Firsov implementation. However, the original code produced a result with incorrect concavity (e.g., a hump instead of a bowl shape). This document explains the mathematical origin of this error and the necessary correction.

2 Mathematical Analysis

2.1 Sign Convention of the Dynamical Weight

According to the original paper (Eq. 61), the dynamical weight is expressed as an exponential of the pairwise interaction:

$$w_{loc}^{dyn} = \exp \left(\sum_n k_n \sum_{\alpha, \beta} S_\alpha S_\beta P_n(\tau_\alpha - \tau_\beta) \right) = \exp \left(\sum_n k_n \alpha_n \right) \quad (1)$$

where $\alpha_n = \sum_{\alpha, \beta} S_\alpha S_\beta P_n(\tau_\alpha - \tau_\beta)$. In Eq. 64 of the paper, the derivative of the free energy $F = -\ln Z$ with respect to the coefficients k_n is stated as:

$$\frac{\partial F}{\partial k_n} = \langle \alpha_n \rangle \quad (2)$$

However, using the standard path integral definition $Z = \int \dots w_{loc}^{dyn}$, the derivative should be:

$$\frac{\partial F}{\partial k_n} = -\frac{1}{Z} \frac{\partial Z}{\partial k_n} = -\langle \alpha_n \rangle \quad (3)$$

This missing minus sign in Eq. (2) propagates into the subsequent extraction of the correlation coefficients q_n .

2.2 Derivation of the Correlation Coefficients

Using the chain rule, the derivative of the free energy with respect to the retarded interaction coefficients d_n is:

$$\frac{\partial F}{\partial d_n} = \sum_p \frac{\partial F}{\partial k_p} \frac{\partial k_p}{\partial d_n} \quad (4)$$

Since $K(\tau)$ is related to the retarded interaction $D(\tau)$ by $K''(\tau) = D(\tau)$, we have the mapping $k_p = \sum_n M_{pn} d_n$, which implies $\frac{\partial k_p}{\partial d_n} = M_{pn}$. Thus:

$$\frac{\partial F}{\partial d_n} = -\sum_p M_{pn} \langle \alpha_p \rangle \quad (5)$$

On the other hand, from the source action $S_{int} = -\frac{1}{2} \iint n(\tau) D(\tau - \tau') n(\tau') d\tau d\tau'$, taking the derivative with respect to d_n gives:

$$\frac{\partial F}{\partial d_n} = \frac{1}{2} \iint \langle n(\tau) n(\tau') \rangle P_n(x_{\tau-\tau'}) d\tau d\tau' = \frac{\beta^2}{2(2n+1)} q_n \quad (6)$$

Equating the two expressions for $\frac{\partial F}{\partial d_n}$ yields the corrected formula for q_n :

$$q_n = -\frac{2n+1}{\beta^2} \sum_p M_{pn}(2\langle\tilde{\alpha}_p\rangle) = -\frac{2n+1}{\beta^2} \sum_p M_{pn}\langle\alpha_p^{\text{code}}\rangle \quad (7)$$

where we have absorbed the factor of 2 into α_p^{code} which sums over all pairs (including $i > j$).

3 The Code Correction

The original implementation in `D0_corr.cpp` missed the global minus sign derived in Eq. (7), computing instead $q_n = +\frac{2n+1}{\beta^2} \sum_p M_{pn}\langle\alpha_p^{\text{code}}\rangle$.

To fix the concavity, the coefficient computation in `collect_results()` is modified to include the missing minus sign:

```
// In measures/D0_corr.cpp
q_n(a, b, n) = -sum * (2.0 * n + 1.0) / (beta * beta);
```

4 Note on the q_0 Coefficient and Static Shift

It is worth noting that integration by parts $\iint n'n'P_n = -\iint nnP_n''$ fundamentally loses information about the average density (the constant of integration). As a result, the reconstruction matrix M_{pn} forces Dirichlet-like boundary conditions on the integrated kernel, meaning the formula above computes the *fluctuating* part of the correlation function. The true q_0 mode should contain the static density-density correlation $\langle n_a n_b \rangle$:

$$q_0^{\text{true}} = q_0^{\text{dynamic}} + \langle n_a n_b \rangle = -\frac{1}{\beta^2} \sum_p M_{p0}\langle\alpha_p^{\text{code}}\rangle + \langle n_a n_b \rangle \quad (8)$$

While the added minus sign perfectly restores the correct positive q_2 mode (and thus the bowl-shaped concavity), a fully quantitative absolute shift in $Q(\tau)$ requires ensuring the correct static expectation value $\langle n_a n_b \rangle$ is added to q_0 .